Quiz 3 – 概率论与数理统计 2021 秋季（姚 - 周四）

考试时长：50 分钟。请将所有答案写在本试卷上。

本人郑重承诺将秉承诚信原则，自觉遵守考场纪律，并承担违纪或作弊带来的后果。

学号：________________　　　　　　姓名：________________

$= \begin{cases} x & 0<x<1 \\ 2-x & 1<x<2 \\ 0 \end{cases}$

$\int_0^1 x^2\,dx + \int_1^2 2x - x^2\,dx$
$= \frac{1}{3}x^3\Big|_{x=0}^{1} + (x^2 - \frac{1}{3}x^3)\Big|_{x=1}^{2}$
$= \frac{1}{3} + 4 - \frac{8}{3} - (1-\frac{1}{3}) = 3 + \frac{1-8+1}{3}$
-2

1、（每小题 4 分）

(1) 设随机变量X的概率密度为$f(x) = \begin{cases} 1-|1-x|, 0<x<2 \\ 0, 其它 \end{cases}$，则$E(X) =$ __1__.

(2) 设随机变量X的概率密度为$f(x) = \begin{cases} kx^a, 0<x<1 \\ 0, 其它 \end{cases}$，其中$k, a>0$，且$E(X) = 0.75$，则$a =$ __2__，$k =$ __3__.

$\int_0^1 kx^a\,dx = 1$　$k\frac{1}{a+1}x^{a+1}\Big|_{x=0}^{1} = \frac{k}{a+1}$　$\int_0^1 kx^{a+1}\,dx = \frac{3}{4}$　$\frac{k}{a+2} = \frac{3}{4}$　$k = a+1$　$k+1 = a+2$　$\frac{k}{k+1} = \frac{3}{4}$　$4k = 3k+3$

$\sum_{k=0}^{+\infty} \frac{C}{2^k k!} = 1$

$\sum_{k=0}^{+\infty} \frac{(\frac{1}{2})^k}{k!} e^{-\frac{1}{2}} = 1$

$\Rightarrow P(X=k) = \frac{1}{2^k \cdot k!} e^{-\frac{1}{2}}$

$X \sim \text{Poisson}(\frac{1}{2})$

$D(X) = \frac{1}{2}$

$D(Y) = 4D(X) = 2$

(3) 设随机变量X的频率函数为$P(X=k) = \frac{C}{2^k k!}, k = 0,1,2,\cdots$，其中$C$为常数，则随机变量$Y = 2X-3$的方差$D(Y) =$ __2__.

~~Poisson(1/2)~~

$E(X)=1$　$D(X) = 2 = EX^2 - 1$

$= E[(XY)^2] - [E(XY)]^2 = E(X^2)E(Y^2) - E(X)^2E(Y)^2$
$= 3\times 3 - 1\times 1$

(4) 设随机变量X与Y相互独立，均服从正态分布$N(1,2)$，则$D(XY) =$ __8__.

(5) 设X_1, X_2, X_3相互独立，且均服从参数为λ的泊松分布，令$Y = \frac{1}{3}(X_1+X_2+X_3)$，则$Y^2$的数学期望为 __$\lambda^2 + \frac{1}{3}\lambda$__

$X \sim EXP(\lambda)$
$E(X) = \lambda$
$D(X) = \lambda$　$EX^2 = \lambda^2 + \lambda$

$E\left[\frac{1}{9}(X_1+X_2+X_3)^2\right] = \frac{1}{9}E[X_1^2 + X_2^2 + X_3^2 + 2(X_1X_2 + X_1X_3 + X_2X_3)]$
$= \frac{1}{3}[E(X_1^2) + 2E(X_1X_2)]$
$= \frac{1}{3}\times(\lambda^2 + \lambda + 2\lambda^2)$
$= \lambda^2 + \frac{1}{3}\lambda$

2、（10 分）设随机变量X的密度函数为　$f_X(x) = \begin{cases} 2x, & 0<x<1, \\ 0, & 其它. \end{cases}$　令$Y = X^2$.

（1）求Y的概率密度$f_Y(y)$；　3　（2）计算$\text{Cov}(X,Y)$和相关系数ρ_{XY}.

$g(x) = x^2$ 在(0,1)单个

$h(y) = \sqrt{y}$

$f_Y(y) = \frac{1}{2\sqrt{y}} \cdot f_X(\sqrt{y})$
$= \frac{1}{2\sqrt{y}} \cdot 2\sqrt{y}$
$= 1 \quad (0<y<1)$

$Cov(X,Y) = E(XY) - E(X)E(Y)$
$E(X) = \int_0^1 2x^2\,dx = \frac{2}{3}x^3\Big|_{x=0}^{1} = \frac{2}{3}$　1
$E(Y) = \int_0^1 y\,dy = \frac{1}{2}y^2\Big|_{y=0}^{1} = \frac{1}{2}$　1
$E(XY) = E(X^3) = \int_0^1 2x^4\,dx = \frac{2}{5}$　1
$\Rightarrow Cov(X,Y) = \frac{2}{5} - \frac{1}{3} = \frac{1}{15}$　1
$E(X^2) = \int_0^1 2x^3\,dx = \frac{1}{2}$　$D(X) = \frac{1}{2} - \frac{4}{9} = \frac{1}{18}$　1
$E(Y^2) = \int_0^1 y^2\,dy = \frac{1}{3}$　$D(Y) = \frac{1}{3} - \frac{1}{4} = \frac{1}{12}$　1
$\rho_{XY} = \frac{\frac{1}{15}}{\frac{1}{\sqrt{18}\sqrt{12}}} = \frac{6\sqrt{6}}{15} = \frac{2}{5}\sqrt{6}$　1

3、（10 分）设X, Y是随机变量，在以下 2 种情况下判断X和Y是否独立？是否相关？并给出理由.

(1) $X \sim U(0,1), Y = X^2$；　(2)　$X \sim U(-1,1), Y = X^2$.

(1) $f_X(x) = 1$　$x \in (0,1)$
~~$f_Y(y) = \frac{1}{2\sqrt{y}}$　$y \in (0,1)$~~
$E(X) = \int_0^1 x\,dx = \frac{1}{2}$　1
$E(Y) = E(X^2) = \int_0^1 x^2\,dx = \frac{1}{3}$　1
$E(XY) = E(X^3) = \int_0^1 x^3\,dx = \frac{1}{4}$　1
$Cov(X,Y) = \frac{1}{4} - \frac{1}{6} = \frac{1}{12}$
相关　1
不独立　1

不用算 ρ_{XY}

(2) $f_X(x) = \frac{1}{2}$　$x \in (-1,1)$
~~$f_Y(y) = \frac{1}{2\sqrt{y}} \cdot \frac{1}{2} + \frac{1}{2\sqrt{y}} \cdot \frac{1}{2} = \frac{1}{2\sqrt{y}}$　$y \in (0,1)$~~
$E(X) = \int_{-1}^{1} \frac{1}{2}x\,dx = \frac{1}{4}x^2\Big|_{x=-1}^{1} = 0$　1
$E(Y) = E(X^2) = \int_{-1}^{1} \frac{1}{2}x^2\,dx = \frac{1}{3}$　1
$E(XY) = E(X^3) = \int_{-1}^{1} \frac{1}{2}x^3\,dx = 0$　1
$Cov(X,Y) = E(XY) - E(X)E(Y) = 0$
$\Rightarrow$ 不相关　1
$Y = X^2 \Rightarrow$ 不独立　1

$Cov(X,Y) = 0 \Leftrightarrow \rho_{X,Y} = 0 \Leftrightarrow$ 不相关 $\Leftarrow$ 独立

4、（10 分）设随机变量(X,Y)的联合密度函数为

$$f(x,y)=\begin{cases}Cxy, & 0\le x\le 1, 0<y\le x\\ 0, & \text{其它}\end{cases}$$

(1) 求常数C；8　2

(2) 求X与Y的边际密度，并判断二者是否独立；

(3) 计算$E(X),E(Y),Cov(X,Y)$.

(1) $\int_0^1 dx\int_0^x Cxy\,dy$
$=\int_0^1 Cx\cdot\frac{1}{2}y^2\Big|_{y=0}^{x}dx$
$=\int_0^1 \frac{C}{2}x^3\,dx$
$=\frac{C}{8}x^4\Big|_{x=0}^{1}=\frac{C}{8}$

(2) $f_X(x)=\int_0^x 8xy\,dy=4xy^2\Big|_{y=0}^{x}=4x^3\quad(0\le x\le 1)$　1

$f_Y(y)=\int_y^1 8xy\,dx=4x^2y\Big|_{x=y}^{1}=4y-4y^3\quad(0\le y\le 1)$　1

不独立　$f(x,y)\ne f_X(x)\cdot f_Y(y)$　1

(3) $E(X)=\int_0^1 4x^4\,dx=\frac{4}{5}x^5\Big|_{x=0}^{1}=\frac{4}{5}$　1　或 $\iint_{0\le y\le x\le 1} x\cdot f(x,y)\,dx\,dy$

$E(Y)=\int_0^1 4y^2-4y^4\,dy=\frac{4}{3}y^3-\frac{4}{5}y^5\Big|_{y=0}^{1}=\frac{4}{3}-\frac{4}{5}=\frac{8}{15}$　1

$E(XY)=\int_0^1dx\int_0^x xy\cdot 8xy\,dy=\int_0^1dx\int_0^x 8x^2y^2\,dy=\int_0^1\frac{8}{3}x^2y^3\Big|_{y=0}^{x}dx=\int_0^1\frac{8}{3}x^5\,dx=\frac{4}{9}x^6\Big|_{x=0}^{1}=\frac{4}{9}$　2

$Cov(X,Y)=\frac{4}{9}-\frac{4}{5}\times\frac{8}{15}=\frac{75\times4-9\times32}{9\times5\times15}=\frac{300-288}{9\times5\times15}=\frac{12}{9\times5\times5}=\frac{4}{225}$　1

5、（10 分）市场上对某种商品的需求量Y是随机变量（单位：吨），它服从区间(2000,4000)上的均匀分布. 假设商店每售出这种商品 1 吨可得利润 3 万元；但假如销售不出而囤积于仓库，则每吨浪费保养费 1 万元. 试求此商店应组织多少货源，才能使利润最大?

$Y\sim U(2000,4000)$

设货源 X 吨，$2000\le X\le 4000$，利润 W 万元

$$W=\begin{cases}3X & Y\ge X\\ 3Y-(X-Y)=4Y-X & Y<X\end{cases}$$

$f_Y(y)=\frac{1}{2000}\quad y\in(2000,4000)$

$E(W\mid X=x)=\int_x^{4000}3x\cdot\frac{1}{2000}dy+\int_{2000}^{x}(4y-x)\cdot\frac{1}{2000}dy$

$=\frac{3x}{2000}\cdot(4000-x)+\frac{2y^2-xy}{2000}\Big|_{y=2000}^{x}$

$=\frac{1}{2000}(12000x-3x^2+x^2-8\times10^6+2000x)$

$=\frac{1}{2000}(-2x^2+14000x-8\times10^6)$

轴　$x=\frac{14000}{4}=3500$

6、（10 分）令$X_1,X_2,\cdots$是独立随机变量序列，具有均值$E(X_i)=\mu_i$和方差$D(X_i)=\sigma_i^2<k$ $(i=1,2,\cdots)$，其中常数$k>0$. 证明：对于任意给定的$\varepsilon>0$，恒有：

$$\lim_{n\to\infty}P\left\{\left|\frac{1}{n}\sum_{i=1}^{n}X_i-\frac{1}{n}\sum_{i=1}^{n}\mu_i\right|<\varepsilon\right\}=1.$$

Chebyshev 不等式：$\forall\varepsilon>0\quad P(|X-E(X)|\ge\varepsilon)\le\frac{D(X)}{\varepsilon^2}$

$\Rightarrow\forall\varepsilon>0\quad P\left(\left|\frac{1}{n}\sum_{i=1}^{n}X_i-E\left(\frac{1}{n}\sum_{i=1}^{n}X_i\right)\right|\ge\varepsilon\right)\le\frac{D\left(\frac{1}{n}\sum_{i=1}^{n}X_i\right)}{\varepsilon^2}$

其中 $E\left(\frac{1}{n}\sum_{i=1}^{n}X_i\right)=\frac{1}{n}\sum_{i=1}^{n}E(X_i)=\frac{1}{n}\sum_{i=1}^{n}\mu_i$

$D\left(\frac{1}{n}\sum_{i=1}^{n}X_i\right)=\frac{1}{n^2}D\left(\sum_{i=1}^{n}X_i\right)\overset{\text{独立}}{=}\frac{1}{n^2}\sum_{i=1}^{n}D(X_i)=\frac{1}{n^2}\cdot\sum_{i=1}^{n}\sigma_i^2<\frac{1}{n^2}\cdot nk=\frac{k}{n}$

$\Rightarrow P\left(\left|\frac{1}{n}\sum_{i=1}^{n}X_i-\frac{1}{n}\sum_{i=1}^{n}\mu_i\right|\ge\varepsilon\right)\le\frac{k}{n\varepsilon^2}$

$\Rightarrow\lim_{n\to\infty}P(\quad\sim\quad\ge\varepsilon)=0$

$\Rightarrow\lim_{n\to\infty}P(\quad\sim\quad<\varepsilon)=1$